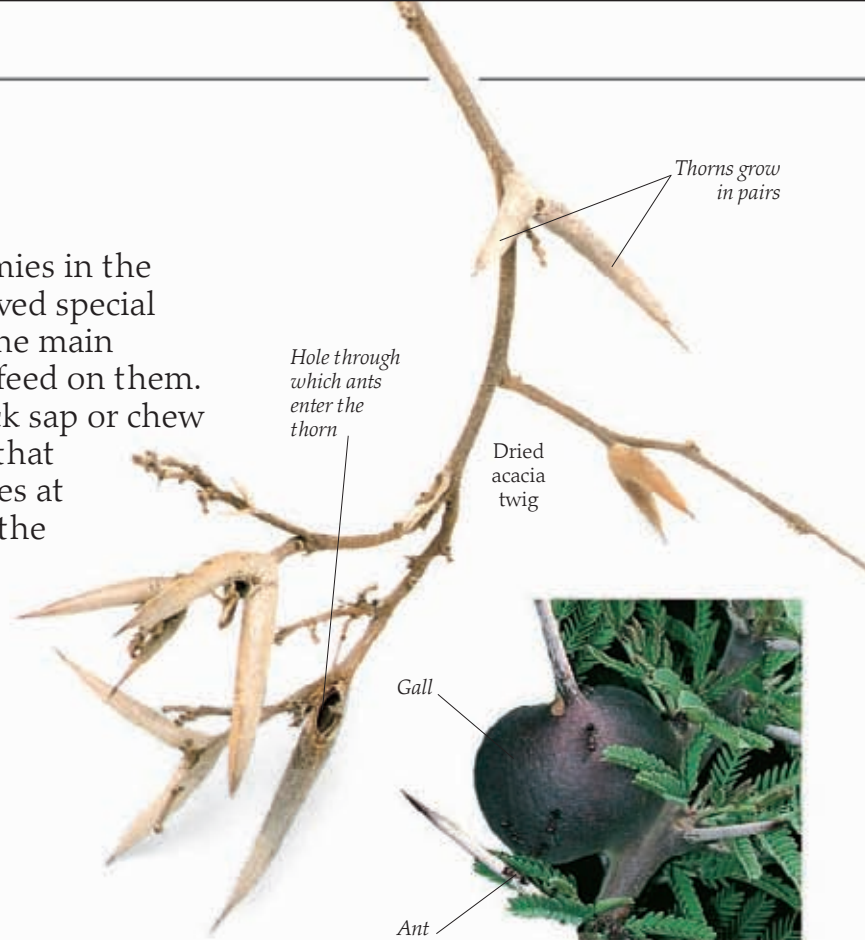


Self-defense

PLANTS CANNOT RUN AWAY from their enemies in the same way that animals do, so they have evolved special weapons and armor to protect themselves. The main enemies of most plants are the animals that feed on them. These range in size from tiny insects that suck sap or chew their way through leaves, to large mammals that eat entire plants. To keep the smallest enemies at bay, many plants have a mat of fine hairs on the surface of their leaves. Larger animals are deterred by means of special weaponry, including spines, thorns, and stings.

As a final defense, many plants have chemicals in their cells that make them unpleasant to eat. Once an animal has tasted the plant, it is unlikely to want to repeat the experience.



ANTS ON GUARD

Some acacia trees rely on ants to keep away browsing animals. In return for food and lodgings, the ants ferociously attack any animal that tries to feed on the tree's leaves. The ants eat the sweet-tasting pith of the thorns and make them hollow. They also feed from a row of nectaries at the base of each leaf. The bullhorn acacia even produces little knobs of protein and fat at the tip of each leaflet, which insures that the ants protect the whole leaf.

CHEMICAL WARFARE

The stings of the nettle are like hypodermic needles. The walls of the sting cell are impregnated with a glasslike substance called silica (p. 7). When an animal brushes past the nettle, the stings puncture its skin and release a cocktail of chemicals that causes a painful irritation. On future encounters, animals remember the sting and avoid the nettle.

Barbs along the edge of the leaf

Long, sharp spines

Small, fleshy leaves

Sharp tip of a nettle sting, magnified many times



RUNNING INTO TROUBLE
Screw pines are tropical plants that have tough, sword-shaped leaves. These have rows of vicious barbs, not only along their edges, but also along their midribs. The barbs point away from the plant, so any animal trying to get near it runs the risk of being impaled as it pushes forward.

Alluaudia

PROTECTED LEAVES
Plants that live in hot, dry climates need to defend themselves, because their leaves are a tempting source of food and water for animals. Many of them, such as cacti, do this with spines (pp. 52–53). This plant from Madagascar has spines that are longer than its leaves, making it very hard for large animals to reach the leaves.

Barbs along the midrib

Nettle

Leaf of the screw pine